

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 12

DATE : 08-12-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry and Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

Corporate Office

ALLEN CAREER INSTITUTE

“SANKALP”, CP-6, Indra Vihar, Kota (Rajasthan)-324005

☎ +91-744-2757575 ☎ +91-744-2435003 ✉ info@allen.ac.in

Rajkot Centres

Kalawad Road

“ISHAVASYAM”, B/h. Crystal Mall, Opp./O.C. Quarters,
Kalawad Road, Rajkot (Gujarat)-360005

☎ 0281-2562131 ☎ +91-85111-43783 ✉ rajkot@allen.ac.in

Limbudiwadi Centre

Opp. Landmark 2 Apartment, Limbudiwadi
Main Road, Off. Kalawad Road, Rajkot (Gujarat)- 360007

☎ 0281-2450228

www.allen.ac.in

Your Target is to secure Good Rank in NEET 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. The potential difference V and the current i flowing through an instrument in an ac circuit of frequency f are given by $V = 5 \cos \omega t$ volts and $I = 2 \sin \omega t$ amperes (where $\omega = 2\pi f$). The power dissipated in the instrument is
 (1) Zero (2) 10 W
 (3) 5 W (4) 2.5 W
2. If instantaneous current is given by $i = 4 \cos(\omega t + \phi)$ amperes, then the r.m.s. value of current is
 (1) 4 amperes (2) $2\sqrt{2}$ amperes
 (3) $4\sqrt{2}$ amperes (4) Zero amperes
3. In an ac circuit, peak value of voltage is 423 volts. Its effective voltage is
 (1) 400 volts (2) 323 volts
 (3) 300 volts (4) 340 volts
4. In an ac circuit $I = 100 \sin 200 \pi t$. The time required for the current to achieve its peak value will be
 (1) $\frac{1}{100}$ sec (2) $\frac{1}{200}$ sec
 (3) $\frac{1}{300}$ sec (4) $\frac{1}{400}$ sec
5. A generator produces a voltage that is given by $V = 240 \sin 120t$, where t is in seconds. The frequency and r.m.s. voltage are
 (1) 60 Hz and 240 V
 (2) 19 Hz and 120 V
 (3) 19 Hz and 170 V
 (4) 754 Hz and 70 V
6. The peak value of an alternating e.m.f. E is given by $E = E_0 \cos \omega t$ is 10 volts and its frequency is 50 Hz. At time $t = \frac{1}{600}$ sec, the instantaneous e.m.f. is
 (1) 10 V (2) $5\sqrt{3}$ V
 (3) 5 V (4) 1 V
7. In an ac circuit, the instantaneous values of e.m.f. and current are $e = 200 \sin 314 t$ volt and $i = \sin\left(314t + \frac{\pi}{3}\right)$ ampere. The average power consumed in watt is
 (1) 200 (2) 100
 (3) 50 (4) 25
8. An alternating current is given by the equation $i = i_1 \cos \omega t + i_2 \sin \omega t$. The r.m.s. current is given by
 (1) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (2) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
 (3) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (4) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$
9. In an ac circuit, the current is given by $i = 5 \sin\left(100t - \frac{\pi}{2}\right)$ and the ac potential is $V = 200 \sin(100t)$ volt. Then the power consumption is
 (1) 20 watts (2) 40 watts
 (3) 1000 watts (4) 0 watt

SPACE FOR ROUGH WORK

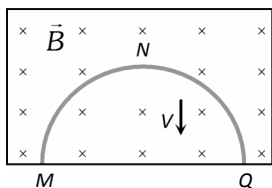
10. A bulb is connected first with dc and then ac of same voltage then it will shine brightly with
 (1) AC
 (2) DC
 (3) Brightness will be in ratio 1/1.4
 (4) Equally with both
11. An alternating current of frequency ' f ' is flowing in a circuit containing a resistance R and a choke L in series. The impedance of this circuit is
 (1) $R + 2\pi fL$
 (2) $\sqrt{R^2 + 4\pi^2 f^2 L^2}$
 (3) $\sqrt{R^2 + L^2}$
 (4) $\sqrt{R^2 + 2\pi fL}$
12. A resonant ac circuit contains a capacitor of capacitance 10^{-6} F and an inductor of 10^{-4} H. The frequency of electrical oscillations will be
 (1) 10^5 Hz (2) 10 Hz
 (3) $\frac{10^5}{2\pi}$ Hz (4) $\frac{10}{2\pi}$ Hz
13. An alternating voltage is connected in series with a resistance R and an inductance L . If the potential drop across the resistance is 200 V and across the inductance is 150 V, then the applied voltage is
 (1) 350 V (2) 250 V
 (3) 500 V (4) 300 V
14. An inductive circuit contains resistance of 10Ω and an inductance of 20 H. If an ac voltage of 120 V and frequency 60 Hz is applied to this circuit, the current would be nearly
 (1) 0.32 amp (2) 0.016 amp
 (3) 0.48 amp (4) 0.80 amp
15. Same current is flowing in two alternating circuits. The first circuit contains only inductance and the other contains only a capacitor. If the frequency of the e.m.f. of ac is increased, the effect on the value of the current will be
 (1) Increases in the first circuit and decreases in the other
 (2) Increases in both the circuits
 (3) Decreases in both the circuits
 (4) Decreases in the first circuit and increases in the other
16. The power factor of an ac circuit having resistance (R) and inductance (L) connected in series and an angular velocity ω is
 (1) $R/\omega L$ (2) $R/(R^2 + \omega^2 L^2)^{1/2}$
 (3) $\omega L/R$ (4) $R/(R^2 - \omega^2 L^2)^{1/2}$
17. A copper rod of length l is rotated about one end perpendicular to the magnetic field B with constant angular velocity ω . The induced e.m.f. between the two ends is
 (1) $\frac{1}{2} B\omega l^2$ (2) $\frac{3}{4} B\omega l^2$
 (3) $B\omega l^2$ (4) $2B\omega l^2$

SPACE FOR ROUGH WORK

18. An e.m.f. of 15 volt is applied in a circuit containing 5 henry inductance and 10 ohm resistance. The ratio of the currents at time $t = \infty$ and at $t = 1$ second is

(1) $\frac{e^{1/2}}{e^{1/2} - 1}$ (2) $\frac{e^2}{e^2 - 1}$
(3) $1 - e^{-1}$ (4) e^{-1}

19. A thin semicircular conducting ring of radius R is falling with its plane vertical in a horizontal magnetic induction B . At the position MNQ , the speed of the ring is V and the potential difference developed across the ring is

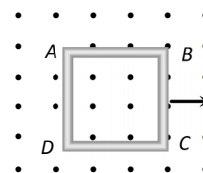


- (1) Zero
(2) $Bv\pi R^2/2$ and M is at higher potential
(3) πRBV and Q is at higher potential
(4) $2RBV$ and Q is at higher potential
20. Two coils have a mutual inductance 0.005 H. The current changes in the first coil according to equation $I = I_0 \sin \omega t$, where $I_0 = 10A$ and $\omega = 100 \pi$ radian/sec. The maximum value of e.m.f. in the second coil is
- (1) 2π (2) 5π (3) π (4) 4π
21. A wire of length 1 m is moving at a speed of 2ms^{-1} perpendicular to its length and a homogeneous magnetic field of 0.5 T. The ends of the wire are joined to a circuit of resistance 6Ω . The rate at which work is being done to keep the wire moving at constant speed is

(1) $\frac{1}{12} W$ (2) $\frac{1}{6} W$

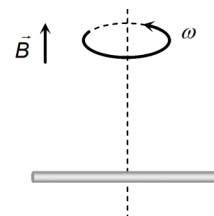
(3) $\frac{1}{3} W$ (4) $1W$

22. A metallic square loop ABCD is moving in its own plane with velocity v in a uniform magnetic field perpendicular to its plane as shown in the figure. An electric field is induced



- (1) In AD, but not in BC
(2) In BC, but not in AD
(3) Neither in AD nor in BC
(4) In both AD and BC

23. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The e.m.f. induced between two ends of the rod is



(1) $B\omega l^2$ (2) $\frac{1}{2} B\omega l^2$
(3) $\frac{1}{8} B\omega l^2$ (4) Zero

SPACE FOR ROUGH WORK

24. The armature of dc motor has 20Ω resistance. It draws current of 1.5 ampere when run by 220 volts dc supply. The value of back e.m.f. induced in it will be
(1) 150 V (2) 170 V
(3) 180 V (4) 190 V
25. A transformer connected to 220 volt line shows an output of 2 A at 11000 volt. The efficiency is 100%. The current drawn from the line is
(1) 100 A (2) 200 A
(3) 22 A (4) 11 A
26. In electromagnetic induction, the induced e.m.f. in a coil is independent of
(1) Change in the flux
(2) Time
(3) Resistance of the circuit
(4) None of the above
27. In electromagnetic induction, the induced charge in a coil is independent of
(1) Change in the flux
(2) Time
(3) Resistance in the circuit
(4) None of the above
28. A coil having an area A_0 is placed in a magnetic field which changes from B_0 to $4B_0$ in a time interval t . The e.m.f. induced in the coil will be
(1) $\frac{3A_0B_0}{t}$ (2) $\frac{4A_0B_0}{t}$
(3) $\frac{3B_0}{A_0t}$ (4) $\frac{4B_0}{A_0t}$
29. In a coil of area 10 cm^2 and 10 turns with a magnetic field directed perpendicular to the plane and is changing at the rate of 10^8 gauss/second. The resistance of the coil is 20 ohm. The current in the coil will be
(1) 5 amp
(2) 0.5 amp
(3) 0.05 amp
(4) 5×10^8 amp
30. A coil of area 100 cm^2 has 500 turns. Magnetic field of 0.1 weber/metre² is perpendicular to the coil. The field is reduced to zero in 0.1 second. The induced e.m.f. in the coil is
(1) 1 V (2) 5 V
(3) 50 V (4) Zero
31. A coil of 100 turns and area 5 square centimetre is placed in a magnetic field $B = 0.2$ T. The normal to the plane of the coil makes an angle of 60° with the direction of the magnetic field. The magnetic flux linked with the coil is
(1) 5×10^{-3} Wb
(2) 5×10^{-5} Wb
(3) 10^{-2} Wb
(4) 10^{-4} Wb
32. In a circuit with a coil of resistance 2 ohms, the magnetic flux changes from 2.0 Wb to 10.0 Wb in 0.2 second. The charge that flows in the coil during this time is
(1) 5.0 coulomb
(2) 4.0 coulomb
(3) 1.0 coulomb
(4) 0.8 coulomb

SPACE FOR ROUGH WORK

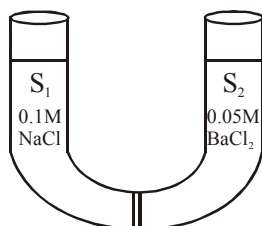
- 33.** Magnetic flux ϕ (in weber) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as $\phi = 5t^2 - 4t + 1$. The induced electromotive force in the circuit at $t = 0.2$ sec. is
- 0.4 volts
 - 0.4 volts
 - 2.0 volts
 - 2.0 volts
- 34.** Lenz's law is expressed by the following formula (here e = induced e.m.f., ϕ = magnetic flux in one turn and N = number of turns)
- $e = -\phi \frac{dN}{dt}$
 - $e = -N \frac{d\phi}{dt}$
 - $e = -\frac{d}{dt} \left(\frac{\phi}{N} \right)$
 - $e = N \frac{d\phi}{dt}$
- 35.** In the case of bar magnet, lines of magnetic induction
- Start from the north pole and end at the south pole
 - Run continuously through the bar and outside
 - Emerge in circular paths from the middle of the bar
 - Are produced only at the north pole like rays of light from a bulb
- 36.** If a bar magnet of magnetic moment M is freely suspended in a uniform magnetic field of strength B , the work done in rotating the magnet through an angle θ is
- $MB(1 - \sin\theta)$
 - $MB \sin \theta$
 - $MB \cos \theta$
 - $MB (1 - \cos \theta)$
- 37.** If the magnetic flux is expressed in weber, then magnetic induction can be expressed in
- Weber/m²
 - Weber/m
 - Weber-m
 - Weber-m²
- 38.** A magnetic needle lying parallel to a magnetic field requires W units of work to turn it through 60° . The torque required to maintain the needle in this position will be
- $\sqrt{3} W$
 - W
 - $\frac{\sqrt{3}}{2} W$
 - $2W$
- 39.** A long magnetic needle of length $2L$, magnetic moment M and pole strength m units is broken into two pieces at the middle. The magnetic moment and pole strength of each piece will be
- $\frac{M}{2}, \frac{m}{2}$
 - $M, \frac{m}{2}$
 - $\frac{M}{2}, m$
 - M, m

SPACE FOR ROUGH WORK

40. Two identical thin bar magnets each of length l and pole strength m are placed at right angle to each other with north pole of one touching south pole of the other. Magnetic moment of the system is
- ml
 - $2ml$
 - $\sqrt{2}ml$
 - $\frac{1}{2}ml$
41. Magnetic lines of force
- Always intersect
 - Are always closed
 - Tend to crowd far away from the poles of magnet
 - Do not pass through vacuum
42. The direction of line of magnetic field of bar magnet is
- From south pole to north pole
 - From north pole to south pole
 - Across the bar magnet
 - From south pole to north pole inside the magnet and from north pole to south pole outside the magnet
43. The magnetic field to a small magnetic dipole of magnetic moment M , at distance r from the centre on the equatorial line is given by (in M.K.S. system)
- $\frac{\mu_0}{4\pi} \times \frac{M}{r^2}$
 - $\frac{\mu_0}{4\pi} \times \frac{M}{r^3}$
 - $\frac{\mu_0}{4\pi} \times \frac{2M}{r^2}$
 - $\frac{\mu_0}{4\pi} \times \frac{2M}{r^3}$
44. A magnet of magnetic moment $50\hat{i} \text{ A-m}^2$ is placed along the x-axis in a magnetic field $\vec{B} = (0.5\hat{i} + 3.0\hat{j})\text{T}$. The torque acting on the magnet is
- $175 \hat{k} \text{ N-m}$
 - $150 \hat{k} \text{ N-m}$
 - $75 \hat{k} \text{ N-m}$
 - $25\sqrt{37} \hat{k} \text{ N-m}$
45. If a magnet is freely suspended in air then it will stop in :-
- Magnetic meridian
 - Geometric meridian
 - Angle of dip
 - None of these

SPACE FOR ROUGH WORK

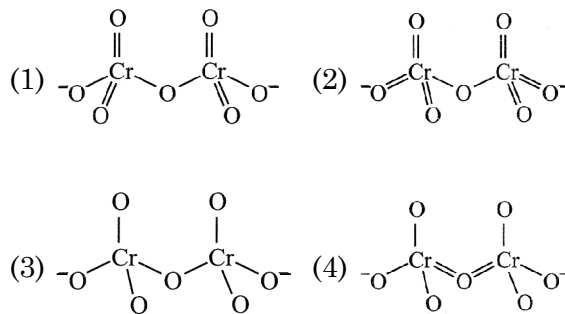
46. How many mili moles of urea should be dissolved in 500 gms of water so as to get a solution which has a difference of 103.57°C between boiling point and freezing point of solution :-
($K_f = 1.86 \text{ K Kg mol}^{-1}$, $K_b = 0.52 \text{ K Kg mol}^{-1}$)
(1) 500 mili moles (2) 900 milli moles
(3) 750 milli moles (4) 650 milli moles
47. Which of the following has been arranged in order of freezing point :-
(1) $0.05 \text{ M KNO}_3 > 0.04 \text{ M BaCl}_2 > 0.140 \text{ M urea} > 0.075 \text{ M CuSO}_4$
(2) $0.04 \text{ M BaCl}_2 > 0.140 \text{ M urea} > 0.075 \text{ M CuSO}_4 > 0.05 \text{ M KNO}_3$
(3) $0.075 \text{ M CuSO}_4 > 0.140 \text{ M urea} > 0.04 \text{ M BaCl}_2 > 0.05 \text{ M KNO}_3$
(4) $0.075 \text{ M CuSO}_4 > 0.05 \text{ M KNO}_3 > 0.140 \text{ M urea} > 0.04 \text{ M BaCl}_2$
48. The Van't Hoff factor for BaCl_2 at 0.01 M concentration is 1.98. The percentage dissociation of BaCl_2 at this concentration is –
(1) 49 (2) 69 (3) 89 (4) 98
49. Two solutions S_1 and S_2 containing 0.1 M NaCl (aq) and $0.05 \text{ M BaCl}_2 \text{ (aq)}$ are separated by semipermeable membrane. Which among the following statement(s) is/are correct ? (Assume complete dissociation of both the electrolytes)



- (1) S_1 and S_2 are isotonic
(2) S_1 is hypertonic while S_2 is hypotonic
(3) S_1 is hypotonic while S_2 is hypertonic
(4) All the above
50. When 20 g of naphthoic acid ($\text{C}_{11}\text{H}_8\text{O}_2$) is dissolved in 50g of benenze ($K_f = 1.72 \text{ K kg mol}^{-1}$), a freezing point depression of 2K is observed. The van't Hoff factor (i) is :
(1) 0.5 (2) 1.0 (3) 2.0 (4) 3.0
51. If two substances A and B have $P_A^0 : P_B^0 = 1 : 2$ and have mole fraction in solution 1 : 2 then mole fraction of A in vapours
(1) 0.33 (2) 0.25 (3) 0.52 (4) 0.2
52. The mass of solute 'A' (mol mass = 40 gm mol^{-1}) that should be added to 180 gm of pure water in order to lower its vapour pressure to $4/5^{\text{th}}$ of its original value :-
(1) 187.5 gm. (2) 90 gm.
(3) 150 gm. (4) 100 gm.
53. A solution of x moles of sucrose in 100 grams of water freezes at -0.2°C . Calculate the amount of ice separate out when it is cooled upto -0.25°C :-
(1) 18 grams (2) 20 grams
(3) 25 grams (4) 23 grams
54. The molarity of a solution made by mixing 50 ml of conc. H_2SO_4 (36 N) with 50 ml. of water, is :
(1) 36 M (2) 18 M
(3) 9 M (4) 6M

SPACE FOR ROUGH WORK

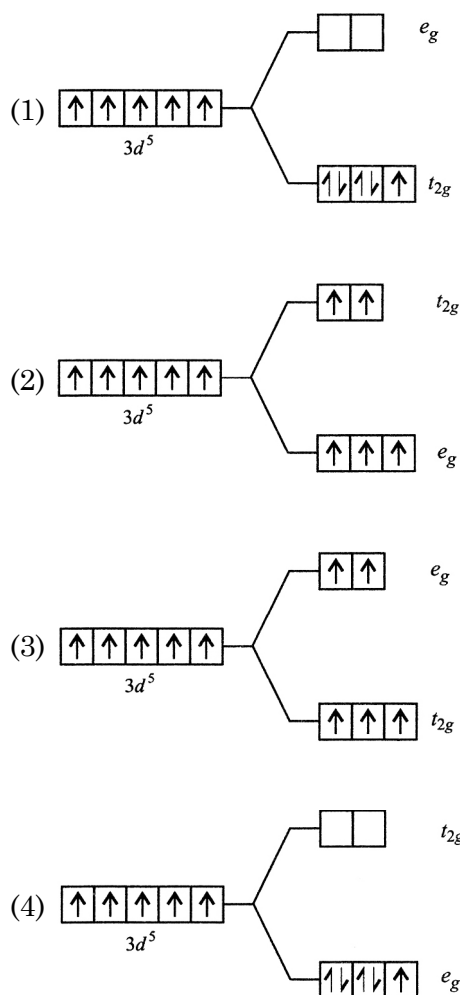
55. Vapour pressure of CCl_4 at 25°C is 143 mm Hg. 0.5 gm of a non-volatile solute (mol. wt. 65) is dissolved in 100 ml of CCl_4 . Find the vapour pressure of the solution. (Density of $\text{CCl}_4 = 1.58 \text{ gm/cm}^3$)
 (1) 141.93 mm (2) 94.39 mm
 (3) 199.34 mm (4) 143.99 mm
56. 150ml of $\text{C}_2\text{H}_5\text{OH}$ (density = 0.78 gm/ml) is diluted to one liter by adding water; molality of the solution is –
 (1) 2.54 (2) 11.7
 (3) 2.99 (4) 29.9
57. pH of a 0.1 M monobasic acid is measured to be 2. Its osmotic pressure at a given temperature T K is :-
 (1) 0.1 RT (2) 0.11 RT
 (3) 1.1 RT (4) 0.01 RT
58. A solution containing 0.1 g of a non-volatile organic substance P (molecular mass 100) in 100g of benzene raises the boiling point of benzene by 0.2°C , while a solution containing 0.1 g of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by 0.4°C . What is the ratio of molecular masses of P and Q ?
 (1) 1 : 2 (2) 2 : 1
 (3) 1 : 4 (4) 4 : 1
59. Two solutions of a substance (non-electrolyte) are mixed in the following manner 480 mL of 1.5M of first solution with 520 mL of 1.2 M of second solution. The molarity of final solution is :-
 (1) 1.20 M (2) 1.50 M
 (3) 1.344 M (4) 2.70 M
60. Just after the slow crystallization of a solvent containing non volatile solute the concentration of solution :-
 (1) Increases (2) Decreases
 (3) Constant (4) None
61. The van't Hoff factor (i) for a dilute solution of $\text{K}_3[\text{Fe}(\text{CN})_6]$ is :-
 (1) 10 (2) 4 (3) 5 (4) 0.25
62. General electronic configuration of transition metals is
 (1) $(n-1)d^{1-10}ns^{0-2}$ (2) $nd^{10}ns^{0-2}$
 (3) $(n-1)d^{10}ns^{0-2}$ (4) $(n-1)d^{1-5}ns^{0-2}$
63. The magnetic moment of a divalent ion in aqueous solution with atomic number 25 is
 (1) 5.9 B.M (2) 2.9 B.M
 (3) 6.9 B.M (4) 9.9 B.M.
64. Colour of transition metal ions are due to absorption of some wavelength. This results in
 (1) d-s transition (2) s-s transition
 (3) s-d transition (4) d-d transition.
65. Identify the correct structure of dichromate ion.



SPACE FOR ROUGH WORK

66. Copper sulphate dissolves in ammonia due to the formation of
- Cu_2O
 - $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 - $[\text{Cu}(\text{NH}_3)_4]\text{OH}$
 - $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4$
67. Mark the correct labelling of different terms used in coordination compounds :-
- $$\begin{array}{c} [\text{Co}(\text{NH}_3)_6]\text{Cl}_3 \\ \text{(iv)} \quad \text{(i)} \quad \text{(iii)} \quad \text{(ii)} \end{array}$$
- (i) Central atom, (ii) Ionisation sphere, (iii) Coordination number, (iv) Ligands
 - (i) Ligands, (ii) Coordination number, (iii) Valency, (iv) Ionisation sphere
 - (i) Ionisation sphere, (ii) Ligands, (iii) Coordination number, (iv) Central atom
 - (i) Ligands, (ii) Ionisation sphere, (iii) Coordination number, (iv) Central atom
68. The formula of the complex diamminechloro-(ethylenediamine)nitroplatinum(IV)chloride is
- $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{en})\text{NO}_2]\text{Cl}_2$
 - $\text{Pt}[\text{Pt}(\text{NH}_3)_2(\text{en})\text{Cl}_2\text{NO}_2]$
 - $\text{Pt}[(\text{NH}_3)_2(\text{en})\text{NO}_2]\text{Cl}_2$
 - $\text{Pt}[(\text{NH}_3)_2(\text{en})\text{NO}_2\text{Cl}_2]$
69. $[\text{Co}(\text{NH}_3)\text{Cl}(\text{en})_2]^{2+}$ shows two geometrical isomers *cis* and *trans*. Which of the following statements is correct?
- trans*-isomer will show optical isomerism.
 - cis*-isomer will show optical isomerism.
 - Both *trans* and *cis*-isomers will show optical isomerism.
 - Neither *cis* nor *trans*-isomer will show optical isomerism.

70. Which of the following energy level diagram for $[\text{FeF}_6]^{3-}$ is correct on the basis of crystal field theory?



71. The tendency to form complexes is maximum for
- Non-metals
 - Transition elements
 - Alkali metals
 - Metals containing fully filled *d*-orbitals.

SPACE FOR ROUGH WORK

- 72.** Which one of the following ions is coloured ?
 (1) Sc^{3+} (2) Ti^{4+}
 (3) Zn^{2+} (4) V^{2+}
- 73.** The atomic radii of transition elements in a row are
 (1) smaller than those of s -block as well as p -block elements
 (2) greater than those of s -block as well as p -block elements
 (3) smaller than those of s -block but greater than those of p -block
 (4) greater than those of s -block but smaller than those of p -block.
- 74.** E^0 values for the couples $\text{Cr}^{3+}/\text{Cr}^{2+}$ and $\text{Mn}^{3+}/\text{Mn}^{2+}$ are -0.41 and $+1.51$ volts respectively. These values suggest that
 (1) Cr^{2+} acts as a reducing agent whereas Mn^{3+} acts as an oxidising agent
 (2) Cr^{2+} is more stable than Cr^{3+} state
 (3) Mn^{3+} is more stable than Mn^{2+}
 (4) Cr^{2+} acts as an oxidising agent whereas Mn^{3+} acts as a reducing agent.
- 75.** When acidified solution of $\text{K}_2\text{Cr}_2\text{O}_7$ is shaken with aqueous solution of FeSO_4 , then
 (1) $\text{Cr}_2\text{O}_7^{2-}$ ion is reduced to Cr^{3+} ions
 (2) $\text{Cr}_2\text{O}_7^{2-}$ ion is converted to CrO_4^{2-} ions
 (3) $\text{Cr}_2\text{O}_7^{2-}$ ion is oxidised to Cr
 (4) $\text{Cr}_2\text{O}_7^{2-}$ ion is oxidised to CrO_3
- 76.** Which of the following statements concerning transition elements is false ?
 (1) They are all metals
 (2) They easily form complex coordination compounds
 (3) Compounds containing their ions are mostly coloured
 (4) They show multiple oxidation states always differing by units of two
- 77.** Which of the following statement is correct ?
 (1) Iron belongs to third transition series of the periodic table
 (2) Iron belongs to f -block of the table
 (3) Iron belongs to first transition series
 (4) Iron belongs to group 7 of the periodic table
- 78.** Which of the following statement is incorrect when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ?
 (1) A deep red vapour is evolved
 (2) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
 (3) Chlorine gas is evolved
 (4) Chromyl chloride is formed.
- 79.** The correct name for the complex $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ is
 (1) Potassium ferric oxalate
 (2) Potassium iron (III) trioxalate
 (3) Potassium trioxalato iron (III)
 (4) Potassium trioxalato ferrate (III).

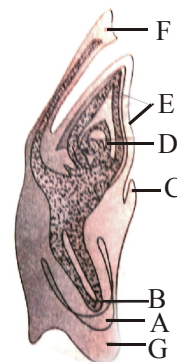
SPACE FOR ROUGH WORK

80. The correct formula of the complex tetraammineaquachlorocobalt (III) chloride is
 (1) $[\text{Cl}(\text{H}_2\text{O})(\text{NH}_3)_4\text{Co}]\text{Cl}$
 (2) $[\text{CoCl}(\text{H}_2\text{O})(\text{NH}_3)_4]\text{Cl}$
 (3) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$
 (4) $[\text{CoCl}(\text{H}_2\text{O})(\text{NH}_3)_4]\text{Cl}$
81. The correct name of the complex ion $[\text{Co}(\text{en})_2\text{Cl}(\text{ONO})]^+$ is
 (1) Cobalt diethylene diamine chloronitrate ion
 (2) Chlorodiethyldiaminenitrito cobalt (III) ion
 (3) Chloronitritodiethyldiamine cobaltate (III) ion
 (4) Chlorobis(ethylenediamine)nitrito cobalt (III) ion.
82. The correct name for the complex $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{C}_2\text{O}_4)_3]$ is
 (1) Hexaamminechromium(III) trioxalatocobalt(III)
 (2) Hexaamminechromate(III) trioxalatocobaltate(III)
 (3) Hexaamminechromium(III) trioxalatocobaltate(III)
 (4) Hexaamminechromate(III) trioxalatocobalt(III).
83. Which of the following is a pair of ionization isomers ?
 (1) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$
 (2) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (3) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ and $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 (4) cis- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and trans- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
84. The complex ions $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$ are called
 (1) Ionization isomers
 (2) Linkage isomers
 (3) Coordination isomers
 (4) Geometrical isomers
85. Which of the following complex or the complex ion will show geometrical isomerism ?
 (1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (2) $[\text{Pt}(\text{NH}_3)\text{Cl}_5]^-$
 (3) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
86. Geometrical isomerism in coordination compounds is exhibited by
 (1) square planar and tetrahedral complexes
 (2) square planar and octahedral complexes
 (3) tetrahedral and octahedral complexes
 (4) square planar, tetrahedral and octahedral complexes
87. The complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is
 (1) Tetrahedral and paramagnetic
 (2) Tetrahedral and diamagnetic
 (3) Square planar and paramagnetic
 (4) Square planar and diamagnetic
88. The hybridization involved in $[\text{Ni}(\text{CO})_4]$ is
 (1) sp^3 (2) dsp^2 (3) sp^3d (4) d^2sp^3
89. Which of the following is an outer orbital complex ?
 (1) $[\text{Fe}(\text{CN})_6]^{3-}$ (2) $[\text{Fe}(\text{CN})_6]^{4-}$
 (3) $[\text{FeF}_6]^{3-}$ (4) $[\text{Cr}(\text{CN})_6]^{3-}$
90. The complex $\text{CoCl}_3 \cdot 6\text{NH}_3$ in aqueous solution ionizes to give a total number of ions equal to
 (1) 0 (2) 2 (3) 3 (4) 4

SPACE FOR ROUGH WORK

91. Read the statements from A to E and answer the question following them.
A – Exine is a continuous layer.
B – Typical microsporangium appears nearly circular in transverse section
C – Endothecium is nutritive layer
D – Pollen mother cell is diploid in nature
E – Pectinase degrades sporopollenin
How many of the above statements are correct
(1) One (2) Two
(3) Three (4) Four
92. The first cell of gametophyte of higher plant is
(1) Pollen grain (2) Megaspore
(3) Spore (4) Zygote
93. In which one pair both the plants can be vegetatively propagated by leaf margin ?
(1) Agave and Kalanchoe
(2) Bryophyllum and Kalanchoe
(3) Asparagus and Bryophyllum
(4) Chrysanthemum and Agave
94. Which of the following propagates through leaf-tip ?
(1) Moss (2) Mint
(3) Marchantia (4) Adiantum
95. Adventive embryony in Citrus is due to
(1) Fertilized egg
(2) Integuments
(3) Zygotic embryo
(4) Nucellus
96. The role of double fertilization in angiosperms is to produce
(1) Cotyledon (2) Pentaploid
(3) Endosperm (4) Integuments
97. Nucellar embryo is
(1) Amphimictic haploid
(2) Amphimictic diploid
(3) Apomictic haploid
(4) Apomictic diploid
98. Both, autogamy and geitonogamy are prevented in
(1) Maize (2) Cucumber
(3) Castor (4) None of these

99. Longitudinal section of an embryo of grass is given. Select the correct option :



	Coleoptile	Epiblast	Radicle	Scutellum
(1)	B	A	F	C
(2)	D	B	A	E
(3)	D	A	C	C
(4)	E	C	B	F

100. Which of the following statement is applicable for all flowering plants ?
(1) Monosiphonous pollen grains
(2) Non-motile and morphologically dissimilar gametes
(3) Presence of pollinium
(4) Division of generative cell after pollination
101. If the chromosome number in the root cell of Zea mays is 20, then what would be the chromosome number in the cell of radicle (a) hypocotyl (b) epiblast (c) aleurone layer (d) and scutellum (e) ?
(1) a = 20 ; b = 30 ; c = 20 ; d = 30 ; e = 20
(2) a = 20 ; b = 20 ; c = 20 ; d = 30 ; e = 20
(3) a = 20 ; b = 30 ; c = 30 ; d = 30 ; e = 20
(4) a = 30 ; b = 20 ; c = 20 ; d = 20 ; e = 30
102. The tapetal cells of anther shows
(1) Haploidy (2) Diploidy
(3) Triploidy (4) Polyploidy
103. Fibrous bands present in endothecium of anther wall is made up of
(1) Sporopollenin (2) α -cellulose
(3) Pectin (4) Lignin
104. Tapetum secretes
(1) Sporopollenin
(2) Pollenkitt
(3) Compatibility proteins
(4) All of these

- 105.** Even after killing the generative cell with a laser beam the pollen grain of a flowering plant germinates and produces normal pollen tube because
- (1) Laser beam stimulates pollen germination and pollen tube growth
 - (2) The vegetative cell has not been damaged
 - (3) The contents of killed generative cell permit germination and pollen tube growth
 - (4) The laser beam does not damage the region from which pollen tube emerges
- 106.** When pollen grains of a flower fall on the stigma of another flower of the same plant this condition is called
- (1) Genetically self pollination
 - (2) Genetically self pollination and ecologically cross pollination
 - (3) Ecologically cross pollination
 - (4) Genertically cross pollination and ecologically self pollination
- 107.** Which is not a contrivance for self pollination?
- (1) Homogamy
 - (2) Bud pollination
 - (3) Cleistogamy
 - (4) Dichogamy
- 108.** Which is generally not a characteristic of anemophilous flower ?
- (1) Unisexual nature
 - (2) Abundant pollen grains
 - (3) Bright coloured
 - (4) Reduction in no of sepals, petals and ovules
- 109.** Dioecious condition and epihydrophyly is found in
- (1) Vallisneria
 - (2) Salvia
 - (3) Zostera
 - (4) Coconut
- 110.** When some natural barriers exist between the stamens and pistil to check self pollination, it is known as
- (1) Heterostyly
 - (2) Dichogamy
 - (3) Herkogamy
 - (4) Dicliny
- 111.** When the hilum, chalaza and micropyle of ovule lie in the same long axis, it is known as
- (1) Amphitropous
 - (2) Anatropous
 - (3) Orthotropous
 - (4) Campylotropous
- 112.** In female gametophyte, which cell is mother cell of endosperm ?
- (1) Synergid
 - (2) Antipodal
 - (3) Central cell
 - (4) Egg
- 113.** Most common type of endosperm is
- (1) Nuclear
 - (2) Cellular
 - (3) Helobial
 - (4) Any of these
- 114.** What will be the ploidy of endosperm and zygote if the cross is made between $6n$ ♀ plant and $4n$ ♂ plant ?
- (1) $5n$, $8n$
 - (2) $8n$, $5n$
 - (3) $10n$, $7n$
 - (4) $8n$, $2n$
- 115.** Development of sporophyte without gametic fusion is known as
- (1) Apomixis
 - (2) Apospory
 - (3) Apogamy
 - (4) All of these
- 116.** In some angiosperms and gymnosperm the multiple embryos are produced in the following way/ways ?
- (1) By fertilization of synergids or antipodal cells by sperms
 - (2) By cleavage of the single zygote
 - (3) One or more cells of the nucellus or integument develops into embryo
 - (4) All of these
- 117.** In the process of asexual reproduction
- (1) Large number of individuals are produced due to involvement of reduction division
 - (2) Individuals as genetically similar to one another but not to their parent
 - (3) There is no need to search for a mate
 - (4) Consumption of resources is very high
- 118.** Grafting method can be used
- (1) In all tracheophytes
 - (2) Only in gymnospermic plants
 - (3) In cambium containing plants
 - (4) Only in monocot plants

- 119.** Which of the following organisms show great synchrony between sexes and release a large number of gametes into the surrounding medium to enhance the chances of fertilisation
- All terrestrial plants
 - Majority of fungi and few algae
 - Majority of algae
 - All spermatophytes
- 120.** Which of the following plants produce non-motile male gametes ?
- Ulothrix, Marchantia
 - Strobilanthus, Chara
 - Spirogyra, Ulothrix
 - Mangifera, Pinus
- 121.** In which of the following plant, grafting is not possible ?
- Plum
 - Pear
 - Mango
 - Maize
- 122.** Select the odd one out w.r.t. life span
- Life spans of organisms are not correlated with their size
 - Crows and parrots show wide difference in their life span
 - Life span of rice is 3-4 weeks
 - It is the period between birth and natural death of an organism
- 123.** Select the correct match w.r.t. vegetative propagules in angiosperms
- Rhizome : Water hyacinth
 - Offset : Bryophyllum
 - Bulbil : Oxalis
 - Leaf buds : Ginger
- 124.** Select the odd one out w.r.t. wall layers of microsporangium in flowering plants
- Endothecium
 - Middle layers
 - Tapetum
 - Hilum
- 125.** Which of the following weed has become a major cause of pollen allergy in India ?
- Pistia
 - Myosotis
 - Parthenium
 - Mirabilis
- 126.** Select the odd one out w.r.t. the pollen grains
- Pollen grains are rich in nutrients
 - Its consumption increases the performance of athletes and race horses
 - It can be stored for years in liquid nitrogen
 - Pollen grains possess non-sticky covering called pollen kitt
- 127.** The function of pollen tube in angiosperms is to carry
- Flagellate sperms
 - Male and female gametes
 - Non-flagellate male gametes
 - Female gametes
- 128.** Polysiphonous condition of pollen grain is found in
- Cucurbitaceae
 - Malvaceae
 - Both (1) & (2)
 - Solanaceae
- 129.** When ovule as well as embryo sac is curved like horse shoe, the type is called as
- Circinotropous
 - Amphitropous
 - Hemianatropous
 - Anatropous
- 130.** Polygonum type of embryo sac is :
- 8-celled
 - 15-nucleated
 - Haploid
 - Exosporic type
- 131.** Mark the odd option (w.r.t. contrivances of autogamy)
- Homogamy
 - Cleistogamy
 - Dicliny
 - Bud pollination
- 132.** Pollination occurs by pseudocopulation mechanism in
- Ophrys
 - Fig
 - Mango
 - Zea mays
- 133.** The asexual reproduction in saccharomyces occurs by means of
- Oidia
 - Conidia
 - Chlamydo spores
 - Budding
- 134.** Which of the following plants is monocarpic ?
- Mangifera
 - Acacia
 - Bambusa
 - Zizyphus
- 135.** Select the correct statement
- Perennials are never polycarpic
 - Carrot is monocarpic annual plant
 - Gymnosperms are perennial
 - Polycarpic plants are mostly annuals and biennials

136. The process by which organisms with different evolutionary history evolve similar phenotypic adaptations in response to a common environmental challenge is called

- (1) Natural selection
- (2) Convergent evolution
- (3) Non-random
- (4) Adaptive radiation

137. The eye of octopus and eyes of cat show different patterns of structure, yet they perform similar function. This is an example of

- (1) homologous organs that have evolved due to convergent evolution
- (2) Homologous organs that have evolved due to divergent evolution
- (3) Analogous organs that have evolved due to convergent evolution
- (4) Analogous organs that have evolved due to divergent evolution

138. Following are the two statements regarding the origin of life

- (1) The earliest organisms that appeared on the earth were non-green and presumably anaerobes
- (2) The first autotrophic organisms were the chemoautotrophs that never released oxygen

Of the above statements which one of the following options is correct ?

- (1) (1) is correct but (2) is false
- (2) (2) is correct but (1) is false
- (3) Both (1) and (2) are correct
- (4) Both (1) and (2) are false

139. Presence of NaCl in body fluid indicates that life originated in :-

- (1) Primitive ocean
- (2) Fresh water lakes
- (3) Sandy soil
- (4) All the above

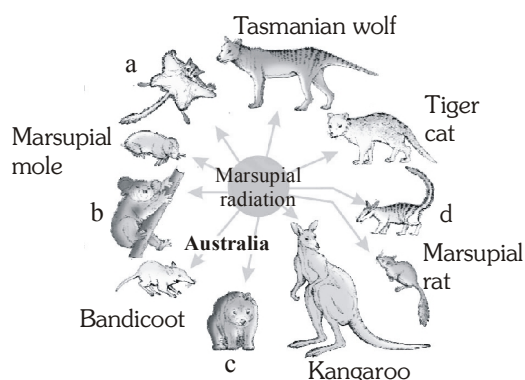
140. Convergent evolution is illustrated by

- (1) Rat and Dog
- (2) Bacterium and Protozoan
- (3) Starfish and Cuttle fish
- (4) Dog fish and Whale

141. Closely related species with same structure and develops in different directions exhibit

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Chemical evolution
- (4) All of the above

142. Recognise the figure and find out the correct matching.

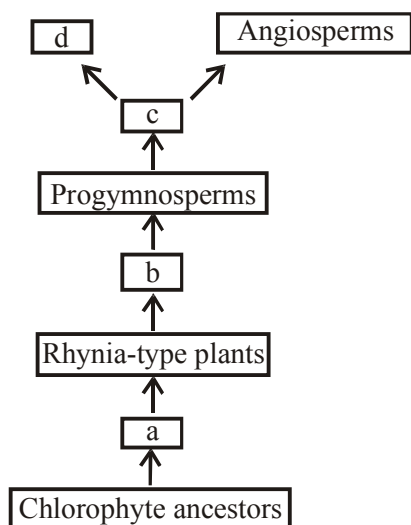


- (1) c-wombat, b-koala, a-sugar glider, d-banded anteater
- (2) a-wombat, c-koala, d-sugar glider, b-banded anteater
- (3) b-wombat, d-koala, c-sugar glider, a-banded anteater
- (4) d-wombat, a-koala, b-sugar glider, c-banded anteater

143. In ...a..., a fish caught in ...b..., happened to be a ...c... which was thought to be extinct. These animals are called ...d... evolved into the first ...e...

- (1) a-1891, b-South America, c-Ichthyosaurs, d-lobefins, e-amphibians
- (2) a-1938, b-South America, c-Ichthyophis, d-coelocanth, e-reptiles
- (3) a-1891, b-North America, c-coelocanth, d-lobefins, e-amphibians
- (4) a-1938, b-South Africa, c-coelocanth, d-lobefins, e-amphibians

144. Recognise the given and find out the correct matching.



- (1) a-psilophyton, b-tracheophyte ancestors, c-cycads, d-conifers
- (2) a-tracheophyte ancestors, b-psilophytons, c-seed ferns, d-cycads
- (3) a-psilophyton, b-tracheophyte ancestors, c-seed ferns, d-cycads
- (4) a-tracheophyte ancestors, b-psilophyton, c-cycads, d-seed ferns

145. Correct order is

- (1) PalaeozoicArchaeozoicCoenozoic
- (2) ArchaeozoicPalaeozoicProterozoic
- (3) PalaeozoicMesozoicCoenozoic
- (4) MesozoicArchaeozoicProterozoic

146. Darwin had made his sea voyage through

- (1) H.M.S. Beagle (2) Titanic
- (3) Carpathia (4) Rocky Mountain

147. Birds and mammals evolved from :-

- (1) Reptiles (2) Frog
- (3) Salamander (4) Coelocanth

148. Formation of protobiont in medium for :-

- (1) Getting isolation in medium
- (2) Acquiring energy from electrode
- (3) Formation of non-colloidal drop
- (4) All of the above

149. Longer chemical reactions produces simple carbonic compounds in oceans.

This statement is related to :-

- (1) Theory of special creation
- (2) Spontaneous generation theory
- (3) Chemogeny
- (4) Cosmozoic theory

150. Which of the following has ability to work as a replicating material as well as a enzyme ?

- (1) RNA
- (2) DNA
- (3) Protobiont
- (4) Coacervate

151. The term "*Prebiotic soup*" was coined by :-

- (1) Oparin
- (2) Haldane
- (3) Aristotle
- (4) Thales

152. According to the idea of Big bang the universe is :-

- (1) Expanding
- (2) Contracting
- (3) Steady
- (4) Both (2) and (3)

153. Arrange the events of chemogeny and biogeny given below in correct order :-

- A. Origin of simple inorganic compounds
- B. Origin of complex organic compounds
- C. Origin of giant molecules
- D. Origin of free atoms
- E. Origin of prebiotic soup
- F. Origin of prokaryotes

- (1) D, A, E, B, C, F
- (2) A, B, E, D, C, F
- (3) D, A, B, E, C, F
- (4) F, C, B, E, A, D

154. Match the following columns and find **correct** combination :

Column I		Column II	
a	Darwin	p	Mutation theory
b	De Vries	q	Protobiont
c	Pasteur	r	Origin of species
d	Fox	s	Special Creation
		t	Swan-Necked Flask Experiment

- (1) a = r, b = p, c = t, d = q
 (2) a = p, b = q, c = r, d = s
 (3) a = t, b = r, c = q, d = p
 (4) a = r, b = t, c = p, d = q

155. Which one is **not correct** regarding the evolution of horses ?

- (1) Increase in their height
 (2) Elongation of their neck
 (3) Reduction of toes of fingers in legs
 (4) Absence of enamel on teeth

156. Who published the book "Origin of Species by Natural Selection" in 1859

- (1) Lamarck (2) Darwin
 (3) Wallace (4) Oparin

157. All of the following arrangements were made by S.L. Miller to create primitive earth like conditions in a laboratory scale, except one :-

- (1) Electric discharge as source of energy.
 (2) Closed flask containing CH_4 , H_2 , NH_3 in a ratio of 2 : 2 : 1.
 (3) Killed yeasts in a sterilised flask.
 (4) Water vapour through boiling water and torrential rain through condensor.

158. How many of the following are examples of connecting link?

- (a) *Peripatus* (b) *Pavo*
 (c) *Neopilina* (d) *Euglena*
 (e) Lung fish
 (1) One (2) Two
 (3) Three (4) Four

159. How many of the following develops as a result of divergent evolution?

- (a) Sweet potato and potato.
 (b) Thorn of bougainvillea & tendril of cucurbita.
 (c) Flippers of penguin and dolphin.
 (d) Wings of bat and flippers of whale.
 (e) Eyes of octopus and mammals.
 (f) Sting of honey bee and scorpion
 (1) Two (2) Three (3) Four (4) Five

160. Mark the option with the correct pair of Australian marsupials (from column-I) and placental mammals (from column-II) showing convergent evolution :-

	Column-I		Column-II
(A)	Numbat	(i)	Anteater
(B)	Spotted cuscus	(ii)	Flying squirrel
(C)	Marsupial mole	(iii)	Lemur
(D)	Tiger cat	(iv)	Mole
		(v)	Bobcat

- (1) A-(i), B-(v), C-(ii), D-(iii)
 (2) A-(i), B-(iii), C-(ii), D-(v)
 (3) A-(iv), B-(iii), C-(ii), D-(v)
 (4) A-(i), B-(iii), C-(iv), D-(v)

161. Which of the following was referred as "golden age of reptile"?

- (1) Mesozoic (2) Palaeozoic
 (3) Coenozoic (4) Azoic

162. The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats). The phenomenon called as :-

- (1) Chemical evolution
 (2) Adaptive radiation
 (3) Convergent evolution
 (4) Adaptive convergence

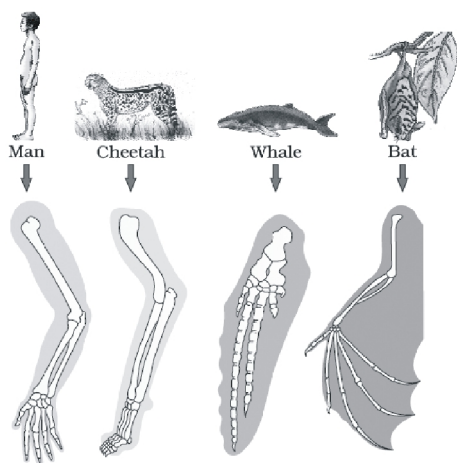
163. "Principles of population" book was written by:-

- (1) Darwin (2) De Vries
 (3) Thales (4) Malthus

164. Which of the following statement is **true** ?

- (1) The geological history of earth closely correlates with the biological history of earth
- (2) The fitness, according to Darwin, refers ultimately and only to reproductive fitness
- (3) Branching descent and natural selection are two key concept of Darwinian theory of evolution
- (4) All statements are true

165. Mark the **incorrect** statement regarding the animals shown in the figure below :-



- (1) It indicates their common ancestry
- (2) It indicates divergent evolution
- (3) It shows homologous organs.
- (4) It shows different structures evolving for the same function

166. Which of the following is a **correct** set of vestigial organs ?

- (1) Coccyx, vermiform appendix, ear
- (2) Nails, nictitating membrane, coccyx
- (3) Ear muscle, patella, caecum
- (4) Auricular muscle, body hair, wisdom tooth

167. In the developmental history of mammalian chambered heart, it is observed that it passes through a two chambered fish - like heart, three - chambered frog - like heart and finally four chambered stage. To which hypothesis this above cited statement can be approximated?

- (1) Biogenetic law
- (2) Hardy - Weinberg law
- (3) Lamarck's principle
- (4) Mendelian principle

168. Who was the first to explain recapitulation theory:

- (1) Weismann
- (2) Muller and Haeckel
- (3) Darwin
- (4) Malthus

169. Analogous organs are :

- (1) Similar in origin
- (2) Similar in structure
- (3) Disimilar in origin
- (4) Disimilar in function

170. Connecting link between annelida and mollusca:

- (1) Cuttle fish
- (2) Octopus
- (3) *Neopilina*
- (4) *Nautilus*

171. *Change with descent* is the basis of which theory :

- (1) Recapitulation theory
- (2) Oparin's theory
- (3) Darwinism
- (4) Cell theory

172. The structure of beak in the following diagram appears as result of :-



- (1) Competition for food
- (2) Adaptation
- (3) Adaptive radiation
- (4) All of the above

173. Darwin was influenced by the observations of :-

- (1) Oparin
- (2) Haldane
- (3) Malthus
- (4) Lamark

174. The invertebrates were formed and active by the time about

- (1) 350 MYA
- (2) 500 MYA
- (3) 200 MYA
- (4) 2000 MYA

175. Select the incorrectly matched ?

- (1) Sea weeds and few plants existed – probably around 320 MYA
- (2) Jawless fish – probably evolved around 350 MYA
- (3) Fish with stout and strong fins could move on land – probably around 350 MYA
- (4) Coelocanth – probably evolved during 500 MYA

176. Select odd with respect to habitat ?

- (1) *Stegosaurus*
- (2) *Brachiosaurus*
- (3) *Ichthyosaurus*
- (4) *Tyrannosaurus*

177. The dianosarus probably disappeared from earth during

- (1) about 500 MYA
- (2) 350 MYA
- (3) 20 MYA
- (4) 65 MYA

178. Read the following statements.

- I. Small sized reptiles of the period of dianosaurs, still exist today
- II. The fossils of first mammals are small sized.
- III. Probably dianosaurs were killed due to climatic changes.
- IV. The early fauna of south America resembles to horse, hippopotamus, bear, rabbit etc.
- V. Pouched mammals of australia survived due to full of competition from any other mammal.

The number of correct statements are.

- (1) One
- (2) Two
- (3) Four
- (4) Five

179. In a famous experiment, Weismann cut off tails of newborn mice generation after generation. At the end of experiment, tails of mice neither disappeared nor shortened. This experiment

- (1) Supported Darwin's theory of natural selection
- (2) proved that the tail is an essential organ of vertebrates
- (3) disproved de Vries mutation theory
- (4) Showed that Lamarck was wrong in his theory of inheritance of acquired characters

180. "Human population grows in geometric ratio while food materials increase in arithmetic proportion." It is a statement of :

- (1) Darwin
- (2) Bateson
- (3) Hugo de Vries
- (4) Malthus



Space For Rough Work

Space For Rough Work

Result : NEET (UG) 2019

ALLEN है... तो मुमकिन है

AIR

1



NALIN KHANDELWAL
Classroom



8 in Top 10 All India Ranks



61 in Top 100 All India Ranks

TOTAL STUDENTS QUALIFIED

Classroom
63415

95102*

Distance
31687

AIR
2



BHAVIK BANSAL
Distance

AIR
4



SWASTIK BHATIA
Distance

AIR
5



ANANT JAIN
Classroom

AIR
6



SARTHAK R. BHAT
Distance

AIR
8



DHRUV KUSHWAHA
Distance

AIR
9



MIHIR RAI
Distance

AIR
10



RAGHAV DUBEY
Classroom

AIR-11



HEMANT KHANDELWAL
Distance

AIR-17



RIDHAM AGARWAL
Distance

AIR-18



HARSHVI N. J.
Classroom

AIR-19



ARUNANGSHU B.
Classroom

AIR-20



SATVIK LODHA
Classroom

AIR-21



HARSH JAIN
Distance

AIR-22



HARDIK GUPTA
Classroom

AIR-24



LALIT AGARWAL
Classroom

AIR-26



APOORV RAGHAV
Classroom

AIR-27



PARTH GILL
Distance

AIR-28



GAUTAM CHELLANI
Distance

AIR-30



HARSH AGARWAL
Classroom

AIR-37



LOHITASHYA GOYAL
Distance

AIR-38



VISHWA RAKESH
Distance

AIR-39



ANURAG GAUTAM
Distance

AIR-44



M. H. SAYAN SHAH
Classroom

AIR-45



SHRIKAR GUPTA
Distance

AIR-46



DIPRASOM DAS
Distance

AIR-47



BAHADUR SINGH
Distance

AIR-48



AMAN AGRAWAL
Classroom

AIR-49



VARUN CHOPRA
Distance

AIR-51



ISHAAN MISHRA
Distance

AIR-52



DISHA S. AGRAWAL
Distance

AIR-53



SARVAGYA JAIN
Distance

AIR-54



HARRY GOYAL
Distance

AIR-56



ISHIKA GUPTA
Distance

AIR-58



SHREYAS AGRAWAL
Classroom

AIR-60



ABHAY MITTAL
Classroom

AIR-61



CHIRAG JORVEKAR
Distance

AIR-63



VINEET HARSH
Distance

AIR-65



VAMSHI K. NEELA
Distance

AIR-66



SHREY MUDGAL
Classroom

AIR-69



PRIVANSHU AGARWAL
Classroom

AIR-70



ISHIKA KAMRA
Classroom

AIR-71



STUTI KHANDWALA
Classroom

AIR-72



NAVEEN SHARMA
Distance

AIR-75



SARTHAK SHARMA
Classroom

AIR-76



VANSHIKA ARORA
Distance

AIR-79



SAIYAM V. BHATT
Classroom

AIR-81



UDITH MOHAN
Distance

AIR-84



PRAJNAPAN BASU
Distance

AIR-85



HARSH RAMOLIYA
Distance

AIR-86



UTKARSH SHARMA
Classroom

AIR-87



NISHTHA
Distance

AIR-89



ARYAN BANSAL
Distance

AIR-91



SANGEETA AGRAWAL
Classroom

AIR-92



YASHDEEP
Classroom

AIR-93



ISHAM GUPTA
Classroom

AIR-94



MEGHA GUPTA
Classroom

AIR-96



GAUTAM KAUSHAL
Classroom

AIR-97



POOJA YADAV
Distance



AIR-99



PRAGYA MITRA
Distance

AIR-100



VIVEK KUMAR
Classroom

Authenticity of Result :
Power of **ALLEN**



Heartiest
Congratulations!

*As Per Result Compulsed so far